

Exercise 1

Assume that $y_1, \dots, y_{200}$ are realizations of independent Gaussian random variables with variance equal to 1 and mean $\beta_1 + \beta_2 \exp\{z_i\}$ for $i = 1, \dots, 120$, and mean $\beta_1 + \beta_3 \exp\{z_i^2\}$ for $i = 121, \dots, 200$; where the z_i are known constants and $(\beta_1, \beta_2, \beta_3)$ are unknown real parameters.

- a) Are the assumptions of a Gaussian linear model satisfied in the above formulation? Motivate the answer.
- b) State the parameter space and sample space.
- c) Express the model in matrix form: $\underline{Y} = X\underline{\beta} + \underline{\varepsilon}$, explicitly stating how $\underline{Y}$, X , $\underline{\beta}$, and $\underline{\varepsilon}$ are defined and their dimensions. Write the distribution of $\underline{Y}$ and $\underline{\varepsilon}$.
- d) Obtain the expression of the matrix $X^T X$ and the vector $X^T y$; state how these elements should be used to obtain the maximum likelihood estimate $\hat{\underline{\beta}}$.
- e) Write the distribution of the maximum likelihood estimator $\hat{\underline{\beta}}(\underline{Y})$.
- f) Let $\underline{e} = \underline{y} - X\hat{\underline{\beta}}$ be the vector of the residuals. State which of the following identities are satisfied and motivate the answer:

$$\sum_{i=1}^{200} e_i = 0$$

$$\sum_{i=1}^{200} e_i z_i = 0$$

$$\sum_{i=1}^{200} e_i z_i^2 = 0$$

$$\sum_{i=1}^{200} e_i \exp\{z_i\} = 0$$

$$\sum_{i=1}^{200} e_i \exp\{z_i^2\} = 0$$

$$\sum_{i=1}^{120} e_i \exp\{z_i\} = 0$$

(hint: read the indices in the sum!)

6

• PARAM SPACE

$\theta = \mathbb{R}^p \cdot \mathbb{R}^t = \mathbb{R}^3 \cdot 1 \checkmark \sigma^2$

• SAMPLE SPACE

$\underline{Y} = \mathbb{R}^n = \mathbb{R}^{200}$

7

$\underline{Y} = \underline{X} \underline{\beta} + \underline{\varepsilon}$

$\underline{Y}$

(200×1)

$\underline{X}$

(200×3)

$\underline{\beta}$

(3×1)

$+$

$\underline{\varepsilon}$

(200×1)

$\underline{Y} \sim N_{120}(\beta_1 + \beta_2 \exp(z_i); 1)$

FOR $i = 1, \dots, 120$

$\underline{Y} \sim N_{80}(\beta_1 + \beta_3 \exp(z_i^2); 1)$

FOR $i = 121, \dots, 200$

$\underline{\varepsilon} \sim N_p(0; I_{200})$

$\rightarrow \underline{Y} \sim N_p(\underline{\mu}; 1)$

$p = 200$

$\underline{X} = \begin{bmatrix} 1 & e^{z_1} & 0 \\ \vdots & \vdots & \vdots \\ 1 & e^{z_{120}} & 0 \\ 0 & e^{z_{121}^2} & \vdots \\ \vdots & \vdots & \vdots \\ 0 & e^{z_{200}^2} & \vdots \end{bmatrix}$

(200×3)

$e^{z_1} \cdot e^{z_1} = e^{2z_1}$

d

$[X^T X] =$

$\begin{bmatrix} 1 & \dots & 1 \\ e^{z_1} & \dots & e^{z_{120}} & 0 & \dots & 0 \\ 0 & \dots & 0 & e^{z_{121}^2} & \dots & e^{z_{200}^2} \end{bmatrix}$

(3×200)

$\cdot$

$\begin{bmatrix} 1 & e^{z_1} & 0 \\ \vdots & \vdots & \vdots \\ 1 & e^{z_{120}} & 0 \\ 0 & e^{z_{121}^2} & \vdots \\ \vdots & \vdots & \vdots \\ 0 & e^{z_{200}^2} & \vdots \end{bmatrix}$

(200×3)

$=$

$\begin{bmatrix} 200 & \sum_{i=1}^{120} e^{2z_i} & \sum_{i=121}^{200} e^{2z_i^2} \\ \sum_{i=1}^{120} e^{2z_i} & \sum_{i=1}^{120} e^{2z_i^2} & 0 \\ \sum_{i=121}^{200} e^{2z_i^2} & 0 & \sum_{i=121}^{200} e^{2z_i^4} \end{bmatrix}$

(3×3)

$[X^T Y]$

$(3 \times 200)(200 \times 1) = (3 \times 1)$

$\begin{bmatrix} 1 & \dots & 1 \\ e^{z_1} & \dots & e^{z_{120}} & 0 & \dots & 0 \\ 0 & \dots & 0 & e^{z_{121}^2} & \dots & e^{z_{200}^2} \end{bmatrix}$

(3×200)

$\cdot$

$\begin{bmatrix} \beta_1 + \beta_2 e^{z_1} \\ \vdots \\ \beta_1 + \beta_2 e^{z_{120}} \\ \beta_1 + \beta_3 e^{z_{121}^2} \\ \vdots \\ \beta_1 + \beta_3 e^{z_{200}^2} \end{bmatrix}$

(200×1)

$\begin{bmatrix} y_1 \\ \vdots \\ y_{200} \end{bmatrix}$

$=$

$\begin{bmatrix} \sum y_i \\ \sum_{i=1}^{120} y_i e^{z_i} \\ \sum_{i=121}^{200} y_i e^{z_i^2} \end{bmatrix}$

(3×1)

$\hat{\underline{\beta}} = (X^T X)^{-1} X^T Y$

e

$\underline{\beta}(\underline{Y}) \sim N_p(\underline{\beta}; \sigma^2)$

$\sigma^2 = 1$

- GAUSSIAN RANDOM VARIABLES
- $\sigma^2 = 1$
- $Y_i \sim N(\beta_1 + \beta_2 \exp\{z_i\}; 1)$ FOR $i = 1, \dots, 120$
- $Y_i \sim N(\beta_1 + \beta_3 \exp\{z_i^2\}; 1)$ FOR $i = 121, \dots, 200$

8

ASSUMPTION GAUSSIAN

- INDEPENDENT $\rightarrow$ GIVEN FROM TEXT
- ORTHOGONALITY $\rightarrow \sigma^2 = 1$ $\checkmark$
- LINEAR

Exercise 2

Given a set of $n = 30$ observations, consider fitting the model $Y_i \sim \text{Bernoulli}(\pi_i)$ where $\text{logit}(\pi_i) = \beta_1 + \beta_2 x_i$, with x_i is a dummy variable taking value 1 for the first 10 observations and 0 otherwise. Fitting this model returns the following output

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.3863	0.5590	2.480	0.01314
x	-2.0794	0.7826	-2.657	0.00788
Null deviance	47.111			
Residual deviance	39.112			

- a) Write the likelihood, log-likelihood and score functions for (β_1, β_2) . Write the fitted model.
- b) Compute the estimate of the probability $\hat{\pi}$ for $x = 0$ and $x = 1$. Obtain the odds for $x = 0$ and $x = 1$ and interpret them. Give an estimate of the odds ratio and interpret it.
- c) Test the hypothesis $H_0 : \beta_2 = -1$ vs $H_1 : \beta_2 < -1$.
- d) What are the two quantities “Null deviance” and “Residual deviance”?

LIKELIHOOD

L(theta) = \prod \pi_i^{y_i} (1-\pi_i)^{1-y_i}

LOG-LIKELIHOOD

l(theta) = log(L(theta)) = \sum y_i log(\pi_i) + (1-y_i) log(1-\pi_i) = \sum y_i log(e^{\beta_1 + \beta_2 x_i} / (1 + e^{\beta_1 + \beta_2 x_i})) + (1-y_i) log(1 / (1 + e^{\beta_1 + \beta_2 x_i})) = \sum y_i (\beta_1 + \beta_2 x_i) - \sum (y_i + (1-y_i)) log(1 + e^{\beta_1 + \beta_2 x_i}) = \sum y_i (\beta_1 + \beta_2 x_i) - \sum log(1 + e^{\beta_1 + \beta_2 x_i})

SCORE FUNCTION

a = \frac{1}{n} \frac{d}{d\beta_1} l(theta) = \frac{1}{n} \frac{d}{d\beta_1} l(theta)

a = \frac{1}{n} \left[\sum y_i (\beta_1 + \beta_2 x_i) - \log(1 + e^{\beta_1 + \beta_2 x_i}) \right] = \sum y_i - \frac{e^{\beta_1 + \beta_2 x_i}}{1 + e^{\beta_1 + \beta_2 x_i}}

b = \sum y_i x_i - \frac{e^{\beta_1 + \beta_2 x_i}}{1 + e^{\beta_1 + \beta_2 x_i}} \cdot x_i

log(\pi_i / (1-\pi_i)) = 1.3863 - 2.0794 x_i

x=0

P(y_i=1 | x_i=0) = e^{\beta_1} / (1 + e^{\beta_1}) \rightarrow P(y_i=1 | x_i=0) = e^{\beta_1} / (1 + e^{\beta_1}) = e^{\beta_1} \rightarrow e^{\hat{\beta}_1} = 4 \text{ For } x_i=0 \text{ we expect 400 success every 100 failure}

P(y_i=0 | x_i=0) = 1 - e^{\beta_1} / (1 + e^{\beta_1}) = 1 / (1 + e^{\beta_1})

x=1

P(y_i=1 | x_i=1) = e^{\beta_1 + \beta_2} / (1 + e^{\beta_1 + \beta_2})

P(y_i=0 | x_i=1) = 1 - e^{\beta_1 + \beta_2} / (1 + e^{\beta_1 + \beta_2}) = 1 / (1 + e^{\beta_1 + \beta_2})

P(y_i=1 | x_i=1) : P(y_i=0 | x_i=1) = \frac{e^{\beta_1 + \beta_2}}{1 + e^{\beta_1 + \beta_2}} : \frac{1}{1 + e^{\beta_1 + \beta_2}} = e^{\beta_1 + \beta_2} \rightarrow e^{\hat{\beta}_1 + \hat{\beta}_2} = 0.5 \text{ For } x_i=1 \text{ we expect 50 success every 100 failure}

ODDS RATIO

\frac{P(y_i=1 | x_i=1)}{P(y_i=0 | x_i=1)} = \frac{e^{\beta_1 + \beta_2}}{e^{\beta_1}} = e^{\beta_2} \rightarrow e^{\hat{\beta}_2} = 0.125 \text{ The odds of } x_i=1 \text{ are equal to the odds of } x_i=0 \text{ multiplied by } 0.125

\frac{P(y_i=1 | x_i=0)}{P(y_i=0 | x_i=0)} = \frac{e^{\beta_1}}{1 + e^{\beta_1}}

6

H0: \beta_2 = -1 \quad Z_{obs}^1 \sim N(0,1)

H1: \beta_2 < -1

Z_{obs}^1 = \frac{\hat{\beta}_2 - \beta_2}{0.7826} = \frac{-2.0794 - (-1)}{0.7826} = -1.3792

\alpha^{obs} = P(|Z| \geq |Z_{obs}^1|) = 1 - P(Z_{obs}^1) \approx 0.08

- Since $\alpha^{obs} < \alpha$ for $\alpha = 0.1$ REJECT H_0 AT THIS CONFIDENCE LEVEL
- Since $\alpha^{obs} > \alpha$ for $\alpha = 0.05, \alpha = 0.01$ (DO NOT REJECT H_0 AT THIS CONFIDENCE LEVEL

- d) D(NULL) = 2 { \tilde{Y}(SAT) - \tilde{Y}(MKE) }
- SATURATED MODEL $\rightarrow$ n COVARIATES
- MKE MODEL $\rightarrow$ NO COVARIATES
- MODEL $\rightarrow$ p COVARIATES

D(RESIDUAL) = 2 { \tilde{Y}(SAT) - \tilde{Y}(MKE) }

Let $(y_1, \dots, y_5)$ and $(y_6, \dots, y_{10})$ be two independent samples from a Poisson distribution of mean $\exp\{\beta_1\}$ and from a Poisson distribution of mean $\exp\{\beta_1 + \beta_2\}$, respectively.

- Formulate an appropriate Poisson regression model for the expected value of Y_i , $i = 1, \dots, 10$.
- Write the log-likelihood function of $\underline{\beta} = (\beta_1, \beta_2)$ and the score function. Find the maximum likelihood estimate of (β_1, β_2) . Finally, obtain the observed information matrix.
- Determine an approximate distribution of the maximum likelihood estimator $\hat{\underline{\beta}}$ of $\underline{\beta} = (\beta_1, \beta_2)$, and an approximate distribution of the maximum likelihood estimator $\hat{\beta}_1$ of β_1 .
- Provide the interpretation of the coefficient β_2 .
- Define the concept of "saturated model" and obtain the expression of maximum of the log-likelihood for this model.

• POISSON DISTRIBUTION

• $Y_i \sim \text{POISSON}(\mu_i)$

• $\eta_i = \beta_1 + \beta_2 x_i$

• ~~$\mu_i = \exp(\eta_i)$~~ $g(\eta_i) = \eta_i$

$$Y_i = \frac{e^{\eta_i} \mu_i^{Y_i}}{Y_i!}$$

$$\begin{cases} \bullet Y_i \sim \text{POIS}(e^{\beta_1}) & \text{for } i = 1, \dots, 5 \\ \bullet Y_i \sim \text{POIS}(e^{\beta_1 + \beta_2}) & \text{for } i = 6, \dots, 10 \end{cases}$$

$$f(\theta) = \frac{e^{-\eta_i} \mu_i^{Y_i}}{Y_i!} \quad \mu_i = e^{\tilde{x}_i^T \theta}$$

LIKELIHOOD $L(\theta)$

$$L(\theta) = \prod \frac{e^{-\eta_i} \mu_i^{Y_i}}{Y_i!}$$

LOG LIKELIHOOD $l(\theta)$

$$\begin{aligned} l(\theta) &= \log(L(\theta)) = \log\left(\prod \frac{e^{-\eta_i} \mu_i^{Y_i}}{Y_i!}\right) = \sum \log\left(\frac{e^{-\eta_i} \mu_i^{Y_i}}{Y_i!}\right) = \sum -\eta_i + Y_i \log(\mu_i) - \log(Y_i!) \\ &= \sum e^{\tilde{x}_i^T \theta} + Y_i \tilde{x}_i^T \theta - \log(Y_i!) \\ &= \sum -e^{\beta_1 + \beta_2 x_i} + Y_i (\beta_1 + \beta_2 x_i) - \log(Y_i!) \end{aligned}$$

SCORE FUNCTION

$$l_x(\theta) = \begin{cases} a = \frac{d}{d\beta_1} l(\theta) = \frac{d}{d\beta_1} \sum -e^{\beta_1 + \beta_2 x_i} + Y_i (\beta_1 + \beta_2 x_i) - \log(Y_i!) = \sum -e^{\beta_1 + \beta_2 x_i} + Y_i \longrightarrow \sum_{i=1}^5 -e^{\beta_1} + \sum_{i=6}^{10} -e^{\beta_1 + \beta_2} + \sum_{i=1}^{10} Y_i = -5e^{\beta_1} - 5e^{\beta_1 + \beta_2} + \sum_{i=1}^{10} Y_i \\ b = \frac{d}{d\beta_2} l(\theta) = \frac{d}{d\beta_2} \sum -e^{\beta_1 + \beta_2 x_i} + Y_i (\beta_1 + \beta_2 x_i) - \log(Y_i!) = \sum -e^{\beta_1 + \beta_2 x_i} \cdot x_i + Y_i x_i \longrightarrow -5e^{\beta_1 + \beta_2} + \sum_{i=6}^{10} Y_i x_i \end{cases}$$

$$l_{xx}(\theta) = \begin{bmatrix} \frac{d l_{x1}(\theta)}{d\beta_1} & \frac{d l_{x1}(\theta)}{d\beta_2} \\ \frac{d l_{x2}(\theta)}{d\beta_1} & \frac{d l_{x2}(\theta)}{d\beta_2} \end{bmatrix} = \begin{bmatrix} -5e^{\beta_1} - 5e^{\beta_1 + \beta_2} & -5e^{\beta_1 + \beta_2} \\ -5e^{\beta_1 + \beta_2} & -5e^{\beta_1 + \beta_2} \end{bmatrix}$$

(x)

• $\hat{\underline{\beta}}(Y) \sim N_p[\underline{\beta}; \tau(\hat{\underline{\beta}})^{-1}]$

• MARGINAL $\hat{\beta}_1(Y) \sim N[\beta_1; \tau(\hat{\underline{\beta}})^{-1}_{(1,1)}]$

• MARGINAL $\hat{\beta}_2(Y) \sim N[\beta_2; \tau(\hat{\underline{\beta}})^{-1}_{(2,2)}]$

(d)

THE LOG-ODDS OF $x_i = 0$ ARE MULTIPLIED BY $\hat{\beta}_2$ TO OBTAIN THE LOG ODDS OF $x_i = 1$

(e)

SATURATED MODEL

• MODEL WITH k COVARIATES

05. MULTIPLEUM-EX2 EXERCISE 3

3 Exercise 3

Among $n = 15$ poisoned lab rats, data about survival time (Y) and antidote/treatment (T) were collected. The researchers consider three kind of treatment: A, B and C, where

Group	n_j	$\bar{y}_j$	s_j^2
A ($j = 1$)	4	0.347	0.2055
B ($j = 2$)	3	3.067	0.4618
C ($j = 3$)	8	1.764	0.5007

Compute the total deviance (total sum of squares - SST) and explain its components.

$h = 15$ TOTAL DEVIANCE

• $\bar{Y} = \frac{1}{h} \sum_{j=1}^3 n_j \bar{y}_j = 1.647$

• $SSE = \sum (n_j - 1) s_j^2 = 2.308$

• $SSR = \sum n_j (\bar{y}_j - \bar{Y})^2 = 12.91$

• $SST = \sum \sum (y_{ij} - \bar{Y})^2 = SSR + SSE = 15.226$

Exercise 3.1

Perform a statistical test to evaluate if the means (of each group) are homogeneous. Specify the system of hypothesis, the test statistic and the p-value.

$$\begin{cases} H_0: \mu_1 = \mu_2 = \mu_3 = 0 \\ H_1: \neg H_0 \end{cases}$$

$$f_{obs} \sim f_{j-1, h-j} \quad f_{obs} = \frac{\frac{SSR}{j-1}}{\frac{SSE}{h-j}} = \frac{\frac{12.91}{3-1}}{\frac{2.308}{15-3}} = 33.56$$

$\alpha^{obs} = P(f_{j-1, h-j} \geq f_{obs}) \approx 1 - \approx 0$ Since $\alpha^{obs} < \alpha$ for $\alpha = 0.1$; $\alpha = 0.05$; $\alpha = 0.01$ we reject H_0 at all confidence level

Exercise 3.2

Specify an appropriate linear regression model. Given the previous data, find the estimates for each coefficient and interpret them.

• $Y \sim N(\beta_1, \sigma^2)$ for $j = 1$

$x_1 = \begin{cases} 1 & \text{if group } j=2 \\ 0 & \text{otherwise} \end{cases}$

• $Y \sim N(\beta_1 + \beta_2, \sigma^2)$ for $j = 2$

• $Y \sim N(\beta_1 + \beta_3, \sigma^2)$ for $j = 3$

$x_2 = \begin{cases} 1 & \text{if group } j=3 \\ 0 & \text{otherwise} \end{cases}$

• $\beta_1 = \mu_1 \longrightarrow \hat{\beta}_1 = 0.347$

• $\beta_1 + \beta_2 = \mu_2 \longrightarrow \hat{\beta}_2 = \mu_2 - \hat{\beta}_1 = 2.72$

• $\beta_1 + \beta_3 = \mu_3 \longrightarrow \hat{\beta}_3 = \mu_3 - \hat{\beta}_1 = 1.417$

$$\longrightarrow Y_i = 0.347 + 2.72x_{1i} + 1.417x_{2i}$$

• The average survival time for group 1 is 0.347

• When keeping everything constant, if I move from group 1 to group 2 the survival time will increase of 2.72 units on average

• When keeping everything constant, if I move from group 1 to group 3, on average, the survival time will increase of 1.417 units

Exercise 3.3

Propose an alternative test for ex. 3.1 specifying the system of hypothesis. Then, compute R^2 of our model.

$$\begin{cases} H_0: \beta_2 = \beta_3 = 0 \\ H_1: \neg H_0 \end{cases}$$

• Null model: $Y_i = \beta_1 + \epsilon_i$

• Full model: $Y_i = \beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i} + \epsilon_i$

$$f_{obs} \sim f_{j-1, h-j} \quad f_{obs} = \frac{\overset{\text{Null Model}}{SSE_N - SSE_F}}{\frac{SSE_F}{h-j}} = \frac{\overset{\text{Full Model}}{SST - SSE_F}}{\frac{SSE_F}{h-j}} = \frac{15.226 - 2.308}{\frac{2.308}{15-3}} = 33.562$$

$\alpha^{obs} = P(f_{j-1, h-j} > f_{obs}) \approx 1 - \approx 0$ Since $\alpha^{obs} < \alpha$ for $\alpha = 0.1$; $\alpha = 0.05$; $\alpha = 0.01$ I reject H_0

$$R^2 = \frac{SSR}{SST} = \frac{SST - SSE}{SST} = \frac{15.226 - 2.308}{15.226} = 0.848$$

OS-MULTIPLEUM-EX2/ EXERCISE 4

4 Exercise 4

Lab *Precise* got some measurements to see whether the tar contents (in milligrams) for three different brands of cigarettes are different. The measurements are showed in the following table.

Sample	Brand A	Brand B	Brand C
1	10.21	11.32	11.60
2	10.25	11.20	11.90
3	10.24	11.40	11.80
4	9.80	10.50	12.30
5	9.77	10.68	12.20
6	9.73	10.90	12.20

Write the equation of the linear regression model. Find and interpret the estimates of regression coefficients.

$\bar{y}_1 = 10$

$\bar{y}_2 = 11$

$\bar{y}_3 = 12$

$\bar{y} = \frac{1}{n} \sum h_j \bar{y}_j = 11$

$\epsilon: N(0, \sigma^2)$

$Y_i = \beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i} + \epsilon_i$

$x_{1i} = \begin{cases} 1 & \text{if group B} \\ 0 & \text{otherwise} \end{cases}$

$x_{2i} = \begin{cases} 1 & \text{if group C} \\ 0 & \text{otherwise} \end{cases}$

$N_1 = \hat{\beta}_1 = 10$

$N_2 = \hat{\beta}_1 + \hat{\beta}_2 \rightarrow \hat{\beta}_2 = 11 - 10 = 1$

$N_3 = \hat{\beta}_1 + \hat{\beta}_3 \rightarrow \hat{\beta}_3 = 12 - 10 = 2$

$\rightarrow Y_i = 10 + 1x_{1i} + 2x_{2i}$

• THE AVERAGE TAR OF GROUP 1 IS 10 mg

• WHEN KEEPING EVERYTHING CONSTANT, IF I MOVE FROM GROUP 1 TO GROUP 2 THE TAR WILL INCREASE ON AVERAGE BY 1 mg

• WHEN KEEPING EVERYTHING CONSTANT, IF I MOVE FROM GROUP 1 TO GROUP 3 ON AVERAGE TAR WILL INCREASE OF 2 mg

Exercise 4.1

Complete the following table.

Sources of variation	D.o.f.	Deviance
Total		
Regression		
Residual		

Perform the statistical test to evaluate if the means (of each group) are homogeneous.

• $SST = SSE + SSR = \sum_{j=1}^J (y_{ij} - \bar{y})^2 = \sum_{j=1}^J (y_{ij} - \bar{y}_j)^2 = \dots = 6.33 + 1.0242 + 6.38 = 13.7342$

• $SSR = \sum h_j (\bar{y}_j - \bar{y})^2 = 6 + 0 + 6 = 12$

• $SSE = \sum (h_j - 1) s_j^2 = SST - SSR = 1.7342$

$\rightarrow \sum \sum (y_{ij} - \bar{y}_j)^2 =$

$\begin{cases} H_0: \mu_A = \mu_B = \mu_C \\ H_1: \neg H_0 \end{cases} \quad f_{obs} \overset{H_0}{\sim} f_{J-1; n-J}$

$f_{obs} = \frac{\frac{SSR}{J-1}}{\frac{SSE}{n-J}} = \frac{\frac{12}{3-1}}{\frac{1.7342}{18-3}} = 65.46$

$\alpha^{obs} = P(f_{J-1, n-J} \geq f_{obs}) \approx 1 - \dots \approx 0$

SINCE $\alpha^{obs} < \alpha$ FOR $\alpha = 0.1; \alpha = 0.05; \alpha = 0.01$ I REJECT H_0
THE MEAN OF THE THREE GROUPS ARE NOT HOMOGENEOUS

1 Exercise 1

Among $n = 31$ cherry trees, data about *Volume*, *Diameter* and *Height* of each tree were collected.

Exercise 1.1

Specify an appropriate regression model for the response variable *Volume* and the related assumptions. Propose a transformation for all variables by thinking about the geometric relationship of our variables.

$n = 31$

- VOLUME V
- DIAMETER d
- HEIGHT h

$V = \pi x^2 h = \frac{\pi}{4} d^2 h$

$\log(V) = \log\left(\frac{\pi}{4}\right) + 2\log(d) + \log(h)$

$Y_i = \beta_1 + \beta_2 X_{1i} + \beta_3 X_{2i} + \epsilon_i; \epsilon_i \overset{w}{\sim} N(0, \sigma^2)$

Exercise 1.2

Knowing that

$$(X^T X)^{-1} = \begin{bmatrix} 96.572 & 3.139 & -24.165 \\ 3.139 & 0.849 & -1.227 \\ -24.165 & -1.227 & 6.310 \end{bmatrix} \quad X^T y = \begin{bmatrix} 101.455 \\ 263.056 \\ 439.896 \end{bmatrix}$$

Find the estimate for β and write the estimated model.

$\underline{\hat{\beta}} = (X^T X)^{-1} \cdot (X^T y) = \begin{bmatrix} -6.64 \\ 2.04 \\ 1.313 \end{bmatrix}$

$\hat{\beta}_1 = -6.64$
 $\hat{\beta}_2 = 2.04$
 $\hat{\beta}_3 = 1.313$

$Y_i = \hat{\beta}_1 + \hat{\beta}_2 X_{1i} + \hat{\beta}_3 X_{2i} + \epsilon_i$
 $Y_i = -6.64 + 2.04 X_{1i} + 1.313 X_{2i}$

$\underline{Y} = \underline{X} \underline{\beta} + \underline{\epsilon}$

$\underline{Y} \quad (31 \times 1)$
 $\underline{X} \quad (31 \times 3)$
 $\underline{\beta} \quad (3 \times 1)$
 $\underline{\epsilon} \quad (31 \times 1)$

$\epsilon \sim N_{31}(0; I_{31})$
 $Y \sim N_{31}(X\beta; I_{31})$
 $\text{rank}(X) = p \quad (\text{Full Rank})$

$\begin{bmatrix} 1 & x_{11} & x_{21} \\ \vdots & \vdots & \vdots \\ 1 & x_{131} & x_{231} \end{bmatrix}$

Exercise 1.3

Knowing the maximum likelihood estimate $\hat{\sigma}^2 = 0.00598$, find the unbiased estimate $\hat{s}^2$.

Then, calculate the estimated variance-covariance matrix of $\hat{\beta}$.

$\hat{\sigma}^2 = 0.00598$

$\hat{s}^2 = \frac{n}{n-p} \sigma^2 = \frac{31}{31-3} 0.00598 = 6.6 \cdot 10^{-3}$

VARIANCE-COVARIANCE MATRIX

$\text{var}(\hat{\underline{\beta}}) = \sigma^2 (X^T X)^{-1} = s^2 (X^T X)^{-1} = 6.6 \cdot 10^{-3} \begin{bmatrix} 96.572 & 3.139 & -24.165 \\ 3.139 & 0.849 & -1.227 \\ -24.165 & -1.227 & 6.310 \end{bmatrix}$

Exercise 1.4
Perform a statistical test to evaluate if the regression coefficient related to the *Diameter* is significant. Specify the system of hypothesis, the test statistic and the p-value. Then, perform another statistical test with the following null hypothesis $H_0: \beta_2 = \beta_3 = 0$ (Restricted model SSE= 8.309).

$Y_i = -6.64 + 2.04 X_{1i} + 1.313 X_{2i}$

$\uparrow$
Diam

$\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases}$

$\epsilon_{obs}^2 \overset{w}{\sim} \epsilon_{n-p}^2$

$\epsilon_{obs}^2 = \frac{\hat{\beta}_2 - \beta_2}{SE(\hat{\beta}_2)} = \frac{2.04 - 0}{\sqrt{6.6 \cdot 10^{-3} \cdot 0.849}} = 27.25$

$\alpha^{obs} = P(|t_{n-p}| > |\epsilon_{obs}^2|) = 2 \min\{P(t_{n-p} < -\epsilon_{obs}^2), P(t_{n-p} > \epsilon_{obs}^2)\} \approx 2(1-w) \approx 0$

SINCE $\alpha^{obs} < \alpha$ FOR $\alpha = 0.1; \alpha = 0.05; \alpha = 0.01$
I REJECT H_0 AT ALL CONFIDENCE LEVEL

$\begin{cases} H_0: \beta_2 = \beta_3 = 0 \\ H_1: \text{no} \end{cases}$

$f_{obs} \overset{w}{\sim} f_{p-p_0; n-p}$

$f_{obs} = \frac{8.309}{3-1} = \frac{8.309}{2}$

$\frac{8.309}{2} > \frac{8.309}{3-1}$

Exercise 2

The titanic dataset is a collection of data about 714 passengers, and the goal is to predict the survival (Survival: 1 if the passenger survived, 0 if they did not) based on some personal characteristics. In particular, here we consider the ticket class (Class: 1 = first, 0 = second or third; dummy), the gender (Gender: man = 1, woman = 0; dummy), and the age (Age, in years). Fitting a logistic regression model in R produces the following summary:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.5003	0.2462	6.09	0.0000
Class	2.0103	0.2479	8.11	0.0000
Gender	-2.5473	0.2017	??	0.0000
Age	-0.0299	0.0074	-4.06	??
Null deviance: 964.52				
Residual deviance: 675.14				

- a) Write the corresponding theoretical model and the expression of the estimated model.
- b) Write the likelihood and log-likelihood function.
- c) Complete the missing values in the table. For Pr(>|z|) of Age, write an approximate value. What variables are statistically significant?
- d) Provide an estimate of the odds for a woman aged 30 with a ticket of first class (denote this individual as "A"). How do you expect this value to change if you consider a person with the same characteristics but aged 31 (denote this individual as "B")?
- e) Provide the interpretation of the coefficient associated with the Class variable. Given the estimate of this coefficient, what is the effect of this covariate on the survival probability?

$Y_i \sim \text{BERNOULLI}(\pi_i)$ $f(\theta) = (\pi_i)^{y_i} (1-\pi_i)^{1-y_i}$ $\pi_i = \frac{e^{\tilde{x}_i^T \beta}}{1 + e^{\tilde{x}_i^T \beta}}$

LIKELIHOOD

$L(\theta) = \prod f(\theta) = \prod (\pi_i)^{y_i} (1-\pi_i)^{1-y_i}$

LOG-LIKELIHOOD

$l(\theta) = \log(L(\theta)) = \log\left(\prod (\pi_i)^{y_i} (1-\pi_i)^{1-y_i}\right) = \sum \log((\pi_i)^{y_i} (1-\pi_i)^{1-y_i}) = \sum y_i \log(\pi_i) + (1-y_i) \log(1-\pi_i) = \sum y_i \log\left(\frac{e^{\tilde{x}_i^T \beta}}{1 + e^{\tilde{x}_i^T \beta}}\right) + (1-y_i) \log\left(\frac{1}{1 + e^{\tilde{x}_i^T \beta}}\right)$
 $= \sum y_i \tilde{x}_i^T \beta - \sum (y_i + (1-y_i)) \log(1 + e^{\tilde{x}_i^T \beta}) = \sum y_i \tilde{x}_i^T \beta - \log(1 + e^{\tilde{x}_i^T \beta})$

c

	Estimate	Std. Error	z value	Pr(> z)
β_0 (Intercept)	1.5003	0.2462	6.09	0.0000
β_2 Class	2.0103	0.2479	8.11	0.0000
β_3 Gender	-2.5473	0.2017	1 ??	0.0000
β_4 Age	-0.0299	0.0074	-4.06	2 ??
Null deviance: 964.52				
Residual deviance: 675.14				

1 $\begin{cases} H_0: \beta_3 = 0 \\ H_1: \beta_3 \neq 0 \end{cases}$ $Z_{obs}^3 \overset{H_0}{\sim} N(0,1)$ $Z_{obs}^3 = \frac{\hat{\beta}_3 - \beta_3}{0.2017} = \frac{-2.5473 - 0}{0.2017} = -12.63$

2 $\begin{cases} H_0: \beta_4 = 0 \\ H_1: \beta_4 \neq 0 \end{cases}$ $Z_{obs}^4 \overset{H_0}{\sim} N(0,1)$ $Z_{obs}^4 = -4.06$

$\alpha_{obs} = P(|Z_3| > |Z_{obs}|) = 2(1 - \Phi(|Z_{obs}^3|)) \approx 0$

At 0.01; 0.05; 0.1 ALL COEFFICIENTS ARE STATISTICALLY SIGNIFICANT SINCE $\alpha_{obs} < \alpha$

d

INDIVIDUAL A	INDIVIDUAL B
• WOMAN	• WOMAN
• 30 yo	• 31 yo
• 1ST CLASS	• 1ST CLASS

$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_1 + \beta_2 D_1 + \beta_3 D_2 + \beta_4 X_3$
 $\pi_i = \frac{e^{\beta_1 + \beta_2 D_1 + \beta_3 D_2 + \beta_4 X_3}}{1 + e^{\beta_1 + \beta_2 D_1 + \beta_3 D_2 + \beta_4 X_3}} = \frac{e^{\beta_1 + \beta_2 + \beta_4 \cdot 30}}{1 + e^{\beta_1 + \beta_2 + \beta_4 \cdot 30}} = 0.93$ (ODDS OF SURVIVAL FOR INDIVIDUAL A)
 $\frac{e^{\beta_1 + \beta_2 + \beta_4 \cdot 31}}{1 + e^{\beta_1 + \beta_2 + \beta_4 \cdot 31}} = 0.929$

$\beta_1 + \beta_2 D_1 + \beta_3 D_2 + \beta_4 X_3 + \epsilon_i$
1 1ST CLASS
0 OTHERWISE
AGE
1 MALE
0 FEMALE

LOG ODDS $\frac{\log\left(\frac{\pi_i}{1-\pi_i}\right)}{\log\left(\frac{\pi_j}{1-\pi_j}\right)} = \frac{\beta_1 + \beta_2 + \beta_4 \cdot 30}{\beta_1 + \beta_2 + \beta_4 \cdot 31} \rightarrow \frac{\pi_i}{1-\pi_i} : \frac{\pi_j}{1-\pi_j} = \frac{e^{\beta_1 + \beta_2 + \beta_4 \cdot 30}}{e^{\beta_1 + \beta_2 + \beta_4 \cdot 31}} = e^{\beta_4} \rightarrow e^{\hat{\beta}_4} = 0.990$
THE ODDS OF SURVIVAL OF PERSON A ARE EQUIV. TO THE ODDS OF PERSON B MULTIPLIED BY 0.990

② $\hat{\beta}_2 = 2.0103$

- WHEN KEEPING EVERYTHING CONSTANT, IF WE MOVE FROM 2ND OR 3RD CLASS TO FIRST CLASS WE INCREASE THE LOG-ODDS OF SURVIVAL BY $\hat{\beta}_2$ (INCREASE ODDS OF $e^{\hat{\beta}_2}$)
- POSITIVE EFFECT SINCE β_2 IS POSITIVE $\rightarrow$ HIGHER PROBABILITY OF SURVIVING FROM 1ST CLASS PASSENGERS

③ $\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 < 0 \end{cases} \quad Z_{obs}^2 \overset{H_0}{\sim} N(0, 1)$

$$Z_{obs}^2 = \frac{\hat{\beta}_2 - \beta_2}{0.2479} = 2.1093 \quad \alpha^{obs} = P(Z_3 > Z_{obs}^2) = \Phi(Z_{obs}^2) > 0.1$$

- SO SINCE $\alpha^{obs} > \alpha$ FOR $\alpha = 0.1$; $\alpha = 0.05$; $\alpha = 0.01$ I DO NOT REJECT H_0
- I PREFER MODEL WITHOUT CLASS WRT MODEL WITH CLASS

④ $D(NULL) = 2\{\tilde{l}(SAT.) - \tilde{l}(NULL)\}$ • SAT MODEL: MODEL WITH n COVARIATES
• NULL MODEL: MODEL WITHOUT COVARIATES

$D(RESIDUAL) = 2\{\tilde{l}(SAT.) - \tilde{l}(MODEL)\}$ • MODEL WITH p COVARIATES

⑤ SIGNIFICANCE MODEL

$$\begin{cases} H_0: \beta_2 = \beta_3 = \beta_4 = 0 \\ H_1: 1_{H_0} \end{cases}$$

$$W_{obs} \sim \chi^2_{p-p_0}$$

$$W_{obs} = 2 \log \left(\frac{\hat{l}(MODEL)}{\hat{l}(NULL)} \right) = 2\{\hat{l}(MODEL) - \tilde{l}(NULL) - \tilde{l}(SAT.) + \tilde{l}(SAT.)\}$$

$$= D(NULL) - D(RES) = 964.52 - 675.14 = 289.38$$

$$\alpha^{obs} = P(W_3 \geq W_{obs}) = 1 - P(W_3 < W_{obs}) \approx 0$$

SINCE $\alpha^{obs} < \alpha$ FOR $\alpha = 0.01$; $\alpha = 0.05$; $\alpha = 0.1$ I REJECT H_0 AT ALL CONFIDENCE LEVEL

$D(MOD)$ IS IN $[0; n-p]$ MODEL IS GOOD
 $\uparrow$
 $714-4$

Exercise 1

To meet competition or cope with economic slowdowns, corporations sometimes undertake a “reduction in force” (RIF), in which substantial numbers of employees are terminated. Federal and various state laws require that employees be treated equally regardless of their age. In particular, employees over the age of 40 years are in a “protected” class, and many allegations of discrimination focus on comparing employees over 40 with their younger coworkers. Here are the data for a recent RIF:

Terminated	Over40: No	Over40: Yes
Yes	17	71
No	564	835

Exercise 1.1

- (a) Choose an appropriate response variable and then a regression model. Justify your answer.
- (b) Further, find the estimated regression model.

Exercise 1.2

Software gives the estimated slope $\hat{\beta}_2 = 1.0371$ and its standard error $SE(\hat{\beta}_2) = 0.2755$. Transform the results to the odds scale. Summarize the results (using confidence intervals) and write a short conclusion.

0.7096329

↓

$\hat{\beta}_2 = 1.0371 \quad SE(\hat{\beta}_2) = 0.2755$

$\alpha = 0.01 \rightarrow 1 - \alpha/2 = 0.995$

ODD SCALE: $\left(e^{\hat{\beta}_2 - SE(\hat{\beta}_2)Z_{1-\alpha/2}}, e^{\hat{\beta}_2 + SE(\hat{\beta}_2)Z_{1-\alpha/2}} \right)$

$= (1.387; 5.735)$

$$\frac{\frac{(Y_i = 1 | X_i = 1)}{(Y_i = 1 | X_i = 0)}}{\frac{(Y_i = 0 | X_i = 1)}{(Y_i = 0 | X_i = 0)}} = \frac{e^{\hat{\beta}_1 + \hat{\beta}_2}}{e^{\hat{\beta}_1}} = e^{\hat{\beta}_2} = 2.221$$

THE ODDS OF BEING FIRED FOR AN OVER 40 ARE 2.22x TO THE ODDS OF BEING FIRED FOR A YOUNGER PERSON

• OUTPUT BINARY $\rightarrow$ LOGIT

• $Y_i \sim \text{Bernoulli} \left(\frac{\pi_i}{1 + \pi_i} \right)$

• $\text{log} \left(\frac{\pi_i}{1 + \pi_i} \right) = \eta_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2}$

• $\eta_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \epsilon_i$

$\left\{ \begin{array}{l} \text{over 40} \\ \text{below 40} \end{array} \right.$

FIND β_1 AND β_2

• CASES

$(Y_i = 1 | X_i = 1) = \frac{e^{\beta_1 + \beta_2}}{1 + e^{\beta_1 + \beta_2}}$

$(Y_i = 0 | X_i = 1) = 1 - \frac{e^{\beta_1 + \beta_2}}{1 + e^{\beta_1 + \beta_2}} = \frac{1}{1 + e^{\beta_1 + \beta_2}}$

$(Y_i = 1 | X_i = 0) = \frac{e^{\beta_1}}{1 + e^{\beta_1}}$

$(Y_i = 0 | X_i = 0) = 1 - \frac{e^{\beta_1}}{1 + e^{\beta_1}} = \frac{1}{1 + e^{\beta_1}}$

$\frac{(Y_i = 1 | X_i = 0)}{(Y_i = 0 | X_i = 0)} = \frac{e^{\beta_1}}{1 + e^{\beta_1}} \cdot \frac{1 + e^{\beta_1}}{1} = e^{\beta_1} \rightarrow e^{\hat{\beta}_1} = \frac{17}{564} \rightarrow \hat{\beta}_1 = -3.50$

$\frac{(Y_i = 1 | X_i = 1)}{(Y_i = 0 | X_i = 1)} = \frac{e^{\beta_1 + \beta_2}}{1 + e^{\beta_1 + \beta_2}} \cdot \frac{1 + e^{\beta_1 + \beta_2}}{1} = e^{\beta_1 + \beta_2} \rightarrow e^{\hat{\beta}_1 + \hat{\beta}_2} = \frac{71}{835} \rightarrow 1.035$

↓

$\text{log} \left(\frac{\pi_i}{1 + \pi_i} \right) = -3.5 + 1.0350 X_i$

2 Exercise 2

The acquisition literature suggests that takeovers occur either due to conflicts between managers and shareholders or to create a new entity that exceeds the sum of its previously separate components. Other research has offered managerial hubris as a third option, but it has not been studied empirically. Recently, some researchers revisited acquisitions over a 10-year period in the Australian financial system. A measure of CEO overconfidence was based on the CEO's level of media exposure, and a measure of dominance was based on the CEO's remuneration relative to the firm's total assets. They then used logistic regression to see whether CEO overconfidence and dominance were positively related to the probability of at least one acquisition in a year. To help isolate the effects of CEO hubris, the model included explanatory variables of firm characteristics and other potentially important factors in the decision to acquire. The following table summarizes the results for the two key explanatory variables:

Covariates	Estimates	SE
Overconfidence	0.0878	0.0402
Dominance	1.5067	0.0057

Exercise 2.1
Write the estimated regression model and interpret the coefficients estimates.

- BINARY OUTPUT: LOGIT
- $\text{odds}\left(\frac{\pi_i}{1-\pi_i}\right) = \eta_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}$
- $Y_i \sim \text{Bernoulli}(\pi_i)$
- $\eta_i = \beta_1 + \beta_2 x_{i1} + \beta_3 x_{i2} = \hat{\beta}_1 + 0.0878 x_{i1} + 1.5067 x_{i2}$
- WHEN KEEPING DOMINANCE CONSTANT, IF WE INCREASE OVERCONFIDENCE BY 1 UNIT WE INCREASE ACQUISITION ODDS BY 0.0878
- WHEN KEEPING OVERCONFIDENCE CONSTANT, IF WE INCREASE DOMINANCE BY 1 UNIT, WE INCREASE ACQUISITION ODDS BY 1.5067

2.2

$\alpha = 0.05$

$$\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases} \quad Z_{obs}^2 \overset{H_0}{\sim} N(0,1)$$

$$Z_{obs}^2 = \frac{0.0878 - 0}{0.0402} = 2.18$$

$$\alpha^{obs} = P(Z_2 \geq Z_{obs}^2) = 2(1 - \mathbb{P}(Z_{obs}^2 | 1)) = 0.06$$

EXERCISE 1

unit	1	2	3	4	5	6	7	8	9	10
x	5.50	4.80	4.70	3.90	4.50	6.20	6.00	5.20	4.70	4.30
y	3.10	2.30	3.00	1.90	2.50	3.70	3.40	2.60	2.80	1.60

unit	11	12	13	14	15	16	17	18	19	20
x	4.90	5.40	5.00	6.30	4.60	4.30	5.00	5.90	4.10	4.70
y	2.00	2.90	2.30	3.20	1.80	1.40	2.00	3.80	2.20	1.50

From the data it is possible to compute the following useful quantities:

$\sum_{i=1}^{20} x_i = 100.00$ $\sum_{i=1}^{20} y_i = 50$ $\sum_{i=1}^{20} x_i^2 = 509.12$ $\sum_{i=1}^{20} y_i^2 = 134.84$ $\sum_{i=1}^{20} x_i y_i = 257.66$

Moreover, from fitting the model we know that

$\text{var}(\hat{\beta}_1(Y)) = 0.7268^2$ $\text{var}(\hat{\beta}_2(Y)) = 0.1440^2$

$\sum_{i=1}^{20} (y_i - \hat{y}_i)^2 = 3.4063$

$n = 20$

$Y_i = \beta_1 + \beta_2 x_i + \epsilon_i$

$E(\epsilon_i) = 0 \quad \forall i = 1, \dots, n$

$\text{VAR}(\epsilon_i) = \sigma^2 > 0$

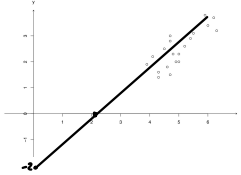
$\text{cov}(\epsilon_i; \epsilon_k) = 0 \quad i \neq k \longrightarrow \text{cov}(Y_i; Y_k) = 0$

MLE OF β_1 AND β_2

$\hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x} \longrightarrow \frac{50}{20} - 0.239 \cdot \frac{100}{20} = -1.69$

$\hat{\beta}_2 = \frac{S_{xy}}{S_{xx}} = \frac{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum x_i y_i - n \bar{x} \bar{y}}{\sum x_i^2 - n \bar{x}^2} = \frac{257.66 - 20 \cdot \frac{100}{20} \cdot \frac{50}{20}}{509.12 - 20 \cdot (\frac{100}{20})^2} = 0.239$

- (WHEN KEEPING EVERYTHING CONSTANT ON AVERAGE, WHEN WE INCREASE THE TEST SCORE OF 1 UNIT WE INCREASE GPA OF 0.239
- ~~NO INTERPRETATION OF $\hat{\beta}_1$ SINCE TEST SCORE OF 0 IS MEANINGLESS~~
- $\hat{\beta}_1$ IS AVERAGE GPA OF TEST SCORE OF 0



$y_i = -1.69 + 0.239 x_i$

$$\begin{cases} y = 0 \\ x = \frac{-1.69}{0.239} = -7.07 \end{cases}$$

CONFIDENCE INTERVAL

$1 - \alpha = 0.99; \alpha = 0.01; \alpha/2 = 0.005; 1 - \alpha/2 = 0.995$

$CI(\hat{\beta}_2) = (\hat{\beta}_2 - t_{n-p, 1-\alpha/2} SE(\hat{\beta}_2); \hat{\beta}_2 + t_{n-p, 1-\alpha/2} SE(\hat{\beta}_2))$

$(-0.239 - 2.578 \cdot 0.144; -0.239 + 2.578 \cdot 0.144)$

$(-0.4245; 0.253)$ NOT INCLUDE 0 IF INCLUDE 0 TEST SCORE x_i MIGHT NOT BE SIGNIFICANT FOR GPA AT 0.01 SIGNIFICANCE LEVEL

EXTRA: FIND $SE(\hat{\beta}_2)$

$$\begin{aligned} \text{VAR}(\hat{\beta}_2) &= \frac{s^2}{\sum (x_i - \bar{x})^2} = \frac{s^2}{\sum x_i^2 - n \bar{x}^2} = \frac{s^2}{509.12 - 20 \cdot (\frac{100}{20})^2} = \frac{0.189}{18} = 0.0207 \\ s^2 &= \frac{SSE}{n-p} = \frac{\sum (y_i - \hat{y}_i)^2}{20-2} = \frac{3.4063}{18} = 0.189 \\ SE(\hat{\beta}_2) &= \sqrt{\text{VAR}(\hat{\beta}_2)} = 0.144 \end{aligned}$$

$\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases}$
 $\alpha = 0.01 \rightarrow$ WE KNOW IT WILL BE STATISTICALLY SIGNIFICANT SINCE 0 IS OUTSIDE THE CI OF 0.99 OF $\hat{\beta}_2$

$t_2^{obs} \stackrel{20-2=18}{\sim} t_{n-p}$
 $t_2^{obs} = \frac{\hat{\beta}_2 - \beta_2}{SE(\hat{\beta}_2)} = \frac{0.839 - 0}{0.149} = 5.626$

$\alpha^{obs} = P(|t_{n-p}| > |t_2^{obs}|) = 2 \min\{P(t_{n-p} < -t_2^{obs}); P(t_{n-p} > t_2^{obs})\} \approx 2(1 - 0.999) \approx 0$

SINCE $\alpha^{obs} < \alpha$ FOR $\alpha = 0.1; \alpha = 0.05; \alpha = 0.01$ I REJECT H_0 AT ALL CONFIDENCE LEVEL β_2 IS SIGNIFICANT (AS EXPECTED)

⑤

$x_A = 5$

$x_B = 6.5$

WIDER INTERVAL FOR STUDENT B
 SINCE x_B IS FURTHER FROM THE AVERAGE $x_A = 5$

SMALLEST INTERVAL FOR $x_A = 5$

1 COMPUTER REPAIR DATA

SIMPLE LINEAR GAUSSIAN MODEL

- $E(\epsilon_i) = 0 \quad \forall i$
- $\text{VAR}(\epsilon_i) = \sigma^2 > 0 \quad \forall i$
- $\text{COV}(\epsilon_i, \epsilon_k) = 0 \quad \text{for } i \neq k \rightarrow \text{COV}(Y_i, Y_k) = 0$
- $\epsilon_i \sim \text{NN}(0, \sigma^2)$
- X_i : FIXED AND NON-STOCHASTIC
- X_i : NOT ALL EQUAL

$$Y_i = \beta_1 + \beta_2 X_i + \epsilon_i$$

$$\bullet \hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x} = \frac{1152}{18} - 14.73 \cdot \frac{21}{18} = -2.322$$

$$\bullet \hat{\beta}_2 = \frac{S_{xy}}{S_{xx}} = \frac{\frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})}{\frac{1}{n-1} \sum (x_i - \bar{x})^2} = \frac{1098}{74.5} = 14.73$$

- $\hat{\beta}_1$ NOT USEFUL SINCE AVERAGE REPAIR TIME FOR 0 MINUTES CALL IS NEGATIVE
- $\hat{\beta}_2$: (WHEN KEEPING EVERYTHING CONSTANT), ON AVERAGE, IF WE INCREASE THE NUMBER OF REPAIRS BY 1 UNIT WE INCREASE TIME OF 14.73 MINUTES

c

$$\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 > 0 \end{cases} \quad \epsilon_{\text{obs}}^2 \overset{18-2=16}{\sim} \chi_{n-p}^2$$

$$t_{\text{obs}}^2 = \frac{\hat{\beta}_2 - \beta_2}{\text{SE}(\hat{\beta}_2)} = \frac{14.73 - 0}{0.51} = 28.316$$

$$\alpha_{\text{obs}}^2 = P(E_{n-p} | \epsilon_{\text{obs}}^2) = 2 \text{NN} \{ P(E_{n-p} < -\epsilon_{\text{obs}}^2); P(E_{n-p} > \epsilon_{\text{obs}}^2) \} \sim (1 - \alpha) \sim 0$$

SINCE $\alpha_{\text{obs}}^2 < \alpha$ FOR $\alpha = 0.1$; $\alpha = 0.05$; $\alpha = 0.01$: REJECT H_0

d SINCE WE REJECTED H_0 AT 5% SIGNIFICANCE LEVEL, WE EXPECT THAT AT 95% CI THERE WILL BE 0 OUTSIDE

$$1 - \alpha = 0.95; \alpha = 0.05; d/2 = 0.025; 1 - \alpha/2 = 0.975$$

$$CI(\hat{\beta}_2) = (\hat{\beta}_2 - \text{SE}(\hat{\beta}_2) \overset{18-2=16}{t_{n-p; 1-\alpha/2}}; \hat{\beta}_2 + \text{SE}(\hat{\beta}_2) \overset{18-2=16}{t_{n-p; 1-\alpha/2}})$$

$$(14.73 - 0.51 \cdot 2.1199; 14.73 + 0.51 \cdot 2.1199)$$

$$(13.64; 15.81)$$

e

$$\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 < 0 \end{cases} \quad \alpha = 0.05$$

$$\epsilon_{\text{obs}}^2 \overset{18}{\sim} \chi_{n-p}^2$$

$$t_{\text{obs}}^2 = \frac{\hat{\beta}_2 - \beta_2}{\text{SE}(\hat{\beta}_2)} = \frac{14.73 - 0}{0.51} = 28.316$$

$$\alpha_{\text{obs}}^2 = P(E_{n-p} | \epsilon_{\text{obs}}^2) = 2 \text{NN} \{ P(E_{n-p} < -\epsilon_{\text{obs}}^2); P(E_{n-p} > \epsilon_{\text{obs}}^2) \} \sim 2(1 - \alpha) \sim 0$$

SINCE $\alpha_{\text{obs}}^2 < \alpha$ FOR $\alpha = 0.05$: REJECT H_0

IS EXPECTED TO BE REJECTED SINCE IF $\beta_2 < 0$ THEN THE TIME DECREASE WHEN NUMBER OF REPAIRS INCREASE

f

$$X^* = 5$$

$$Y^* = \beta_1 + \beta_2 X^* = -2.322 + 14.73 \cdot 5 = 71.328$$

$$Y^* \sim \text{NN}(\beta_1 + \beta_2 X^*; S^2 \left(\frac{1}{n} + \frac{(X^* - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right))$$

$$\frac{\hat{Y}^* - (\beta_1 + \beta_2 X^*)}{\sqrt{S^2 \left(\frac{1}{n} + \frac{(X^* - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right)}} \sim t_{n-p}$$

$$\left(\hat{y}^2 - t_{n-p;1-\alpha/2} \sqrt{s^2 \left(\frac{1}{n} + \frac{(x^2 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right)}; \hat{y}^2 + t_{n-p;1-\alpha/2} \sqrt{s^2 \left(\frac{1}{n} + \frac{(x^2 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right)} \right)$$

$$R^2 = \frac{SSR}{SST} = \frac{SST - SSE}{SST} = \frac{16504 - 32139}{16504} = 0.92$$

Exercise 1

The data contained in the cement dataset represent the hardness (hardness variable) of 13 types of cement with different chemical compositions. Specifically, each type is obtained with varying proportions of aluminium (aluminium variable), silicate (silicate variable), calcium aluminoferrite (aluminium_ferrite), and silicate bic (silicate_bic). The interest is explaining how the hardness of cement depends on the proportions of chemicals.

A regression model was fitted for this purpose and produced the following result:

	Estimate	Std. Error	t statistic	Pr(> t)
β_1 (Intercept)	124.4809	26.7557	4.653	0.0016
β_2 aluminium	0.9739	3.435	0.284	0.0089
β_3 silicate	-0.1405	0.2891	-0.486	0.6400
β_4 aluminium_ferrite	-0.4974	0.2751	-1.807	0.0774
β_5 silicate_bic	0.2751	0.3214	0.856	0.4031

Error sum of squares	49.378
Total sum of squares	2715.763
R^2 coefficient	0.177

- a) Write the model formulation and assumptions.
- b) Complete the missing values in the table. For “Pr(> |t|)” of aluminium_ferrite provide an approximate value. What variables have a statistically significant effect?
- c) Test the statistical hypothesis corresponding to the statement “the covariates do not have an effect on the hardness of cement”.
- d) On a reduced model (“model B”) that includes only the variables aluminium and silicate_bic the error sum of squares is equal to $SSE_B = 74.762$. Perform an F test to compare this model with the complete model (“model A”) that includes all the covariates. Interpret the result: which model would you prefer?
- e) Obtain the coefficient R^2 of model B. Instead of performing the test in point (d), could you have simply compared the coefficient R^2 of the two models? Why?
- f) Figure 1 shows two plots regarding the complete model (model A). Explain what they represent and interpret them.

STATISTICALLY SIGNIFICANT AT 10%.

- $\alpha^{obs} < 0.10$
- INTERCEPT
- ALUMINIUM
- SILICATE-BIC

C COVARIATES NO EFFECT ON HARDNESS OF CEMENT

$$\begin{cases} H_0: \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0 \\ H_1: \text{no} \end{cases} \quad \begin{matrix} p=5 \\ p_0=1 \end{matrix}$$
$$f_{obs} \sim f_{p-p_0; n-p}$$
$$f_{obs} = \frac{\frac{SSE_2 - SSE_1}{p-p_0}}{\frac{SSE_1}{n-p}} = \frac{\frac{2715.763 - 49.378}{5-1}}{\frac{49.378}{13-5}} = 1.08$$

$\alpha^{obs} = P(f_{n-p} > f_{obs}) \approx 1 - 0.9999 \approx 0 \rightarrow$ REJECT H_0 . COVARIATES ARE SIGNIFICANT FOR HARDNESS OF CEMENT

d

MOD A

$Y_i = \beta_1 + \beta_2 x_{i1} + \beta_3 x_{i2} + \beta_4 x_{i3} + \beta_5 x_{i4} + \epsilon_i$

$SSE_A = 49.378$

MOD B: ALUMINIUM, SILICATE BIC

$Y_i = \beta_1 + \beta_2 x_{i1} + \beta_5 x_{i4} + \epsilon_i$

$SSE_B = 74.762$

$$\begin{cases} H_0: \beta_3 = \beta_4 = 0 \\ H_1: \text{no} \end{cases} \quad f_{obs} \sim f_{p-p_0; n-p}$$
$$f_{obs} = \frac{\frac{SSE_B - SSE_A}{p-p_0}}{\frac{SSE_A}{n-p}} = \frac{\frac{74.762 - 49.378}{5-3}}{\frac{49.378}{13-5}} = 2.056$$

$\alpha^{obs} = P(f_{p-p_0, n-p} > f_{obs}) \approx 1 - 0.99 \approx 0.01$ SINCE $\alpha^{obs} > \alpha$ FOR $\alpha = 0.01; \alpha = 0.05; \alpha = 0.1$ I DO NOT REJECT $H_0 \rightarrow$ PREFER THE MODEL WITH ONLY ALUMINIUM AND SILICATE-BIC

- $E(\epsilon_i) = 0$
- $VAR(\epsilon_i) = \sigma^2 > 0$
- $Cov(\epsilon_i; \epsilon_k) = 0$ FOR $i \neq k \rightarrow Cov(Y_i; Y_k) = 0$
- $\epsilon_i \sim NN(0, \sigma^2)$
- VECTORS $x_i \in R^N$ $N=1, \dots, P$ ARE LINEARLY INDEP $\rightarrow RANK(X) = P$ (FULL RANK)
- CONDITIONALLY $Y_{i:1} = x_{i:1}; Y_{i:2} = x_{i:2}$

1 $R^2 = \frac{SSR}{SST} = \frac{SSR - SSE}{SST} = \frac{2715.763 - 49.378}{2715.763} = 0.9817$

2 $\begin{cases} H_0: \beta_5 = 0 \\ H_1: \beta_5 \neq 0 \end{cases} \quad \epsilon_{obs}^{(5)} \sim \epsilon_{n-p} \quad \epsilon_{obs}^{(5)} = \frac{\hat{\beta}_5 - \beta_5}{SE(\hat{\beta}_5)} \rightarrow \hat{\beta}_5 = \epsilon_{obs}^{(5)} \cdot SE(\hat{\beta}_5) = -0.7974$

3 $\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases} \quad \epsilon_{obs}^{(2)} \sim \epsilon_{n-p} \quad \epsilon_{obs}^{(2)} = \frac{\hat{\beta}_2 - \beta_2}{SE(\hat{\beta}_2)} \rightarrow SE(\hat{\beta}_2) = \frac{\hat{\beta}_2 - \beta_2}{\epsilon_{obs}^{(2)}} = 0.2835$

4 $\begin{cases} H_0: \beta_4 = 0 \\ H_1: \beta_4 \neq 0 \end{cases} \quad \epsilon_{obs}^{(4)} \sim \epsilon_{n-p}$

$\epsilon_{obs}^{(4)} = \frac{\hat{\beta}_4 - \beta_4}{SE(\hat{\beta}_4)} = \frac{-0.4974 - 0}{0.2751} = -1.807$

$\alpha^{obs} = P(|\epsilon_{n-p}| > |\epsilon_{obs}^{(4)}|) = 2MIN\{P(\epsilon_{n-p} < -\epsilon_{obs}^{(4)}), P(\epsilon_{n-p} > \epsilon_{obs}^{(4)})\} \approx 2(1 - 0.98) = 0.02$

② $R^2_B = \frac{SSR_B}{SST} = \frac{SST - SSE_B}{SST} = \frac{2715.763 - 74.762}{2715.763} = 0.9724$

CANNOT COMPARE R^2 OF MOD A AND MOD B

• $SSE_A < SSE_B$

• AND SINCE SET B IS Bigger

• THEN $SSR_A > SSR_B$

• AND SO IT IS OBVIOUS THAT $R^2_A > R^2_B$

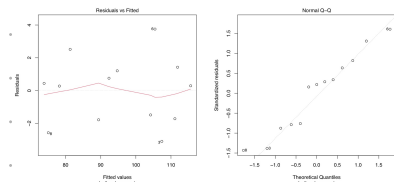


Figure 1:

• RESIDUAL PLOT

• CHECK NORMAL DISTRIBUTION OF ERRORS

• HOMOGENEITY OF ERROR

• NO CORRELATION OF ERROR

• NO PATTERN VISIBLE → GOOD MODEL

• BUT FEW DATA TO MAKE OBSERVATIONS

• FIX PLOT ADD COVARIATES OR TRANSFORMATION

• CHECK ERRORS NORMALITY ASSUMPTION

• FEW OBSERVATION TO MAKE

ASSUMPTION

• IF IN THE LINE → NORMAL DISTRIBUTION OF RESIDUALS

• OUTLIERS IN THE TAIL

• FIX PLOT BY ADDING COVARIATES OR TRANSFORM COVARIATES

Exercise 1

Assume that $y_1, \dots, y_{200}$ are realizations of independent Gaussian random variables with variance equal to 1 and mean $\beta_1 + \beta_2 \exp\{z_i\}$ for $i = 1, \dots, 120$, and mean $\beta_1 + \beta_3 \exp\{z_i^2\}$ for $i = 121, \dots, 200$; where the z_i are known constants and $(\beta_1, \beta_2, \beta_3)$ are unknown real parameters.

- Are the assumptions of a Gaussian linear model satisfied in the above formulation? Motivate the answer.
- State the parameter space and sample space.
- Express the model in matrix form: $\underline{Y} = X\underline{\beta} + \underline{\varepsilon}$, explicitly stating how $\underline{Y}$, X , $\underline{\beta}$, and $\underline{\varepsilon}$ are defined and their dimensions. Write the distribution of $\underline{Y}$ and $\underline{\varepsilon}$.
- Obtain the expression of the matrix $X^T X$ and the vector $X^T \underline{y}$; state how these elements should be used to obtain the maximum likelihood estimate $\hat{\underline{\beta}}$.
- Write the distribution of the maximum likelihood estimator $\hat{\underline{\beta}}(\underline{Y})$.
- Let $\underline{e} = \underline{y} - X\hat{\underline{\beta}}$ be the vector of the residuals. State which of the following identities are satisfied and motivate the answer:

$$\sum_{i=1}^{200} e_i = 0 \quad \sum_{i=1}^{200} e_i z_i = 0 \quad \sum_{i=1}^{200} e_i z_i^2 = 0$$

$$\sum_{i=1}^{200} e_i \exp\{z_i\} = 0 \quad \sum_{i=1}^{200} e_i \exp\{z_i^2\} = 0 \quad \sum_{i=1}^{120} e_i \exp\{z_i\} = 0$$

(hint: read the indices in the sum!)

$$\underline{Y} = X\underline{\beta} + \underline{\varepsilon}$$

$\underline{Y}$ (200x1) = X (200x3) $\underline{\beta}$ (3x1) + $\underline{\varepsilon}$ (200x1)

$\underline{Y} \sim N(\underline{X}\underline{\beta}; 1)$
 $\underline{\varepsilon} \sim N(0; \sigma^2 I_{200})$

$X = \begin{bmatrix} 1 & \exp(z_1) & 0 \\ \vdots & \vdots & \vdots \\ 1 & \exp(z_{120}) & 0 \\ \vdots & \vdots & \vdots \\ 1 & 0 & \exp(z_{121}) \\ \vdots & \vdots & \vdots \\ 1 & 0 & \exp(z_{200}^2) \end{bmatrix}$

$$(X^T X) \quad (3 \times 200)(200 \times 3) \rightarrow (3 \times 3)$$

$$\begin{bmatrix} 1 & \dots & \dots & \dots & \dots & \dots & 1 \\ \exp(z_1) & \dots & \exp(z_{120}) & 0 & \dots & \dots & 0 \\ 0 & \dots & \dots & 0 & \exp(z_{121}^2) & \dots & \exp(z_{200}^2) \end{bmatrix}$$

$$(X^T Y) \quad (3 \times 200)(200 \times 1) = (3 \times 1)$$

$$\begin{bmatrix} 1 & \exp(z_1) & 0 \\ \vdots & \vdots & \vdots \\ 1 & \exp(z_{120}) & 0 \\ \vdots & \vdots & \vdots \\ 1 & 0 & \exp(z_{121}^2) \\ \vdots & \vdots & \vdots \\ 1 & 0 & \exp(z_{200}^2) \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 + \dots + 1 \cdot 1 & \sum_{i=1}^{120} \exp(z_i) & \sum_{i=121}^{200} \exp(z_i^2) \\ \sum_{i=1}^{120} \exp(z_i) & \sum_{i=1}^{120} \exp(z_i^2) & 0 \\ \sum_{i=121}^{200} \exp(z_i^2) & 0 & \sum_{i=121}^{200} \exp(z_i^4) \end{bmatrix}$$

$$\begin{bmatrix} 1 & \dots & \dots & \dots & \dots & \dots & 1 \\ \exp(z_1) & \dots & \exp(z_{120}) & 0 & \dots & \dots & 0 \\ 0 & \dots & \dots & 0 & \exp(z_{121}^2) & \dots & \exp(z_{200}^2) \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_{120} \\ \vdots \\ y_{200} \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{200} y_i \\ \sum_{i=1}^{120} y_i \exp(z_i) \\ \sum_{i=121}^{200} y_i \exp(z_i^2) \end{bmatrix}$$

$$\hat{\underline{\beta}} = (X^T X)^{-1} \cdot X^T \underline{y}$$

$$\hat{\underline{\beta}}(\underline{Y}) \sim N(\underline{\beta}; \sigma^2 (X^T X)^{-1})$$

$p=3$
 1

$$\underline{e} = \underline{y} - X\hat{\underline{\beta}}$$

$$\sum_{i=1}^{200} e_i = 0$$

$$\bullet Y_1, \dots, Y_{120} \quad Y_i \sim N(\beta_1 + \beta_2 \exp z_i; 1)$$

$$\bullet Y_{121}, \dots, Y_{200} \quad Y_i \sim N(\beta_1 + \beta_3 \exp z_i^2; 1)$$

ASSUMPTIONS OF GAUSSIAN LINEAR MODEL

- INDEPENDENT
- HOMOGENEITY $\rightarrow$ SAME VARIANCE
- LINEARITY IN $\beta_1, \beta_2, \beta_3$
- COVARIATES ARE LINEARLY INDEPENDENT
- NORMALITY $\rightarrow$ FULLY NORMAL DISTRIBUTION

$\bullet$ PARAM. SPACE $\Theta = \mathbb{R}^p \mathbb{R}^1 = \mathbb{R}^3$
 $\bullet$ SAMPLE SPACE $\mathcal{Y} = \mathbb{R}^N = \mathbb{R}^{200}$

$$Y_i = \beta_1 + \beta_2 z_i + \beta_3 z_i^2 + \varepsilon_i$$

$$z_i = \begin{cases} z_i & \text{for } i = 1, \dots, 120 \\ 0 & \text{OTHERWISE} \end{cases}$$

$$z_i^2 = \begin{cases} z_i^2 & \text{for } i = 121, \dots, 200 \\ 0 & \text{OTHERWISE} \end{cases}$$